

Sequential Circuits, Latches & Flip-Flops

Memory · SR, D latches · SR, JK, D, T flip-flops · race-around · master-slave · setup, hold, clock-to-Q

QUICK TRICK

This is a big chapter, so learn these four lines first: SR: Set = 1, Reset = 0, 00 = hold, 11 = illegal. JK: 00 hold, 01 reset, 10 set, **11 toggle**. D: Q follows D. T: T = 1 toggles.

11.1 Combinational vs sequential

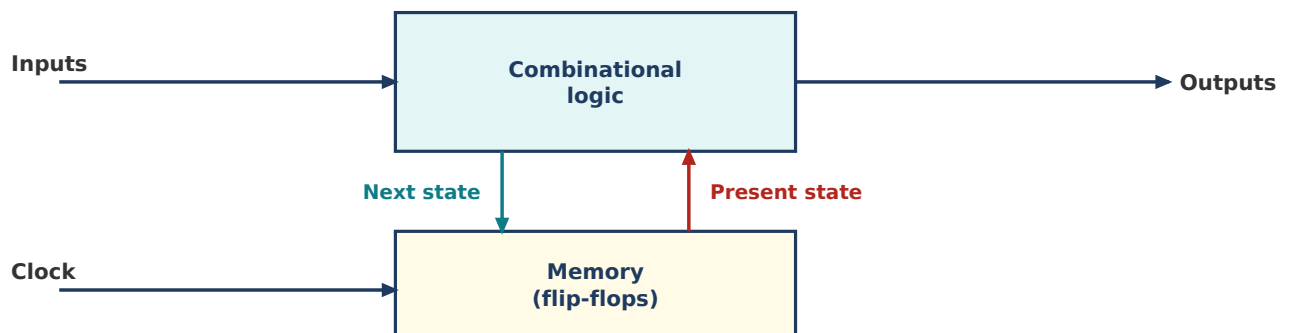
● LEVEL 1 · BEGINNER

Combinational circuit: the output depends only on the **present inputs**. It has no memory. (Adders, MUX, decoders.)

Sequential circuit: the output depends on the **present inputs and on what happened before**. It has **memory**.

Why do we need memory? A computer must remember numbers, counts and the step it is in. Without memory there would be no registers, no counters and no RAM.

Analogy. A calculator with a memory button. Pressing the same key can give different results, because it remembers earlier keys.



Word	Simple meaning
Memory element	A circuit that stores 1 bit (a latch or a flip-flop).
Feedback	The output is sent back to the input. This is how a circuit remembers.
Clock	A regular train of pulses (0, 1, 0, 1...). It tells the circuit when to change.
State	What the memory holds now (for example Q = 1).
Present state / Next state	State now / state after the next clock pulse.
Synchronous	All changes happen together at the clock.
Asynchronous	Changes can happen at any time (no common clock).

EASY MEMORY

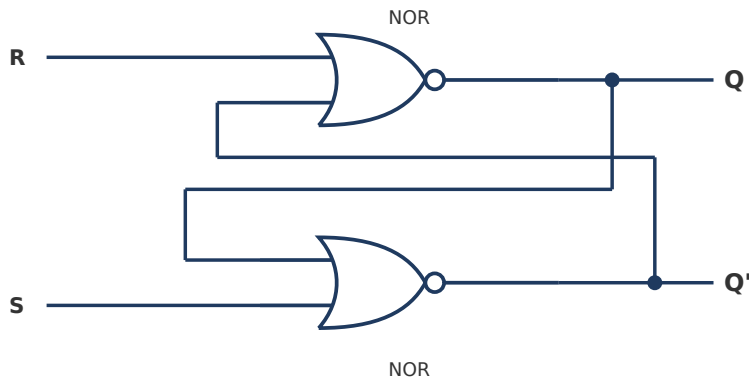
Combinational = **no memory**. Sequential = **memory + feedback** (and usually a clock).

11.2 Latches

● LEVEL 1 • BEGINNER

What is a latch? The simplest 1-bit memory. It is built from two gates that **feed each other** (cross-coupled). A latch is **level-triggered**: it can change while its enable (or its inputs) are at a certain level.

SR latch with NOR gates



S = **Set** (make Q = 1). R = **Reset** (make Q = 0). Q' is the opposite of Q.

S	R	Q (next)	Meaning
0	0	Q (no change)	Hold — remembers the old value
0	1	0	Reset
1	0	1	Set
1	1	Q = Q' = 0	Invalid (and unpredictable when S, R return to 0)

SR latch with NAND gates (inputs are active-LOW)

S'	R'	Q (next)	Meaning
1	1	Q (no change)	Hold
1	0	0	Reset
0	1	1	Set
0	0	Q = Q' = 1	Invalid

QUICK TRICK

Invalid-state trick. The invalid case is when **both inputs are “active” at once**. NOR latch: active = 1, so invalid at **S = R = 1**. NAND latch: active = 0, so invalid at **S' = R' = 0**. The idle (hold) value is 0 for NOR and 1 for NAND.

Gated SR latch and D latch

A **gated latch** has an **enable (EN)** input. When $EN = 0$ the latch ignores S and R. The **D latch** removes the invalid state by using only one input D ($S = D, R = D'$):

EN	D	Q (next)	Meaning
0	X	Q	Hold (latch is closed)
1	0	0	Transparent: Q follows D
1	1	1	Transparent: Q follows D

QUICK TRICK

D latch = “transparent window”. While $EN = 1$, Q copies D (like a glass window). When $EN = 0$ the window is closed and Q keeps the last value.

Latch	Flip-flop
Level -triggered (acts the whole time $EN = 1$)	Edge -triggered (acts only at the clock edge)
Q can change many times while $EN = 1$	Q changes only once per clock pulse
Simpler and faster	Safer for counters and registers
Used inside flip-flops	Built from two latches (master-slave)

11.3 Flip-flops

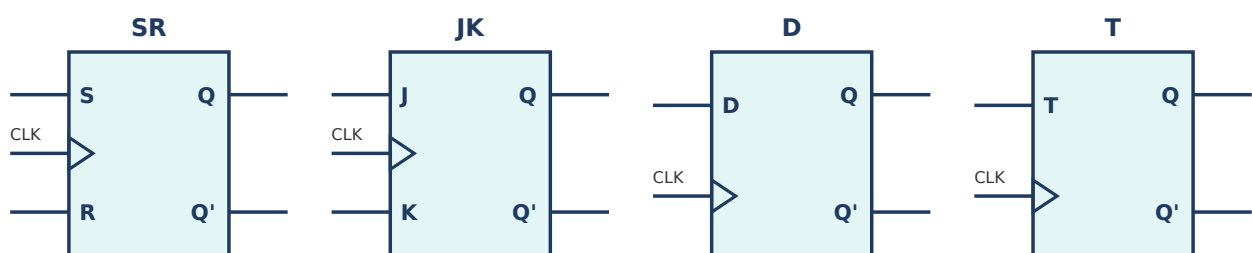
● LEVEL 2 • EXAM

What is a flip-flop? A 1-bit memory that changes **only at a clock edge**. **Why?** It makes all the circuit change together in step, which avoids confusion.

Edge triggering. A **positive (rising) edge** is the moment the clock goes $0 \rightarrow 1$. A **negative (falling) edge** is $1 \rightarrow 0$. The flip-flop looks at its inputs only at that moment.

QUICK TRICK

Symbol trick: a small **triangle** at the clock pin = edge triggered. A triangle with a **bubble** = falling-edge triggered. Triangle only = rising-edge triggered. No triangle = latch or level triggered.



Notation: Q = present state, Q^+ (or Q_{n+1}) = next state after the clock edge.

The four flip-flops — truth table, characteristic table, equation

Characteristic table = the next state Q^+ for every combination of the inputs *and* the present state Q. **Characteristic equation** = the same information written as a formula. X means invalid.

SR flip-flop

Q	S	R	Q+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

JK flip-flop

Q	J	K	Q+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

D flip-flop

Q	D	Q+
0	0	0
0	1	1
1	0	0
1	1	1

T flip-flop

Q	T	Q+
0	0	0
0	1	1
1	0	1
1	1	0

Flip-flop	Characteristic equation	What it does (in words)
SR	$Q^+ = S + R'Q$ (with $SR = 0$)	S sets, R resets, 00 holds, 11 is illegal
JK	$Q^+ = JQ' + K'Q$	00 hold, 01 reset, 10 set, 11 toggle
D	$Q^+ = D$	Q takes the value of D at the clock edge (a 1-clock delay)
T	$Q^+ = T \oplus Q = TQ' + T'Q$	T = 0 hold, T = 1 toggle

QUICK TRICK

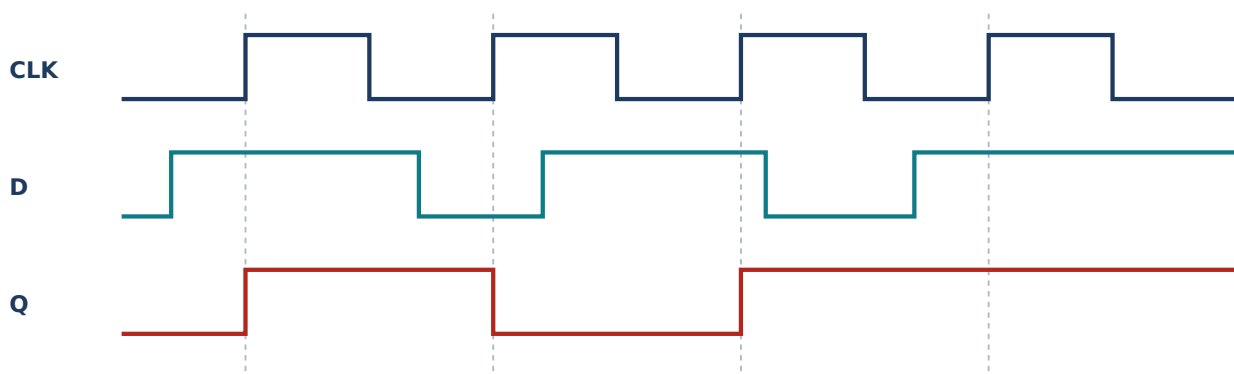
JK is the best friend: J works like S (J = 1 can **set**), K works like R (K = 1 can **reset**), and J = K = 1 does the new thing: **toggle**. That is why JK has no invalid state.

QUICK TRICK

Quick answer for any JK question: read (J, K): **00 → same, 01 → 0, 10 → 1, 11 → opposite**. No need for the equation.

Timing diagrams

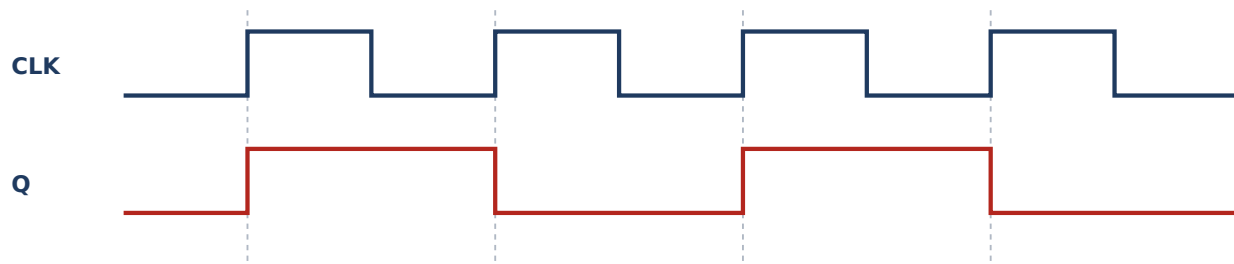
D flip-flop (rising-edge). At each rising edge (dashed line), Q copies the value that D had *just before* the edge. Changes of D between the edges are ignored.



QUICK TRICK

How to read a timing diagram: go to each active clock edge, look at the inputs **just before** the edge, and write the new Q. Q does not change anywhere else.

T flip-flop with $T = 1$. Q toggles at every rising edge, so Q has **half the frequency** of the clock. It is a **divide-by-2** circuit. (This is the idea of counters.)



Excitation tables — “what inputs do I need to get this change?”

An **excitation table** works backwards. You know the present state Q and the state you **want** (Q^+). It tells you which inputs to apply. X = don't care (0 or 1 both work). We need these in counter and state machine design.

Q	Q+	SR: S R	JK: J K	D	T
0	0	0 X	0 X	0	0
0	1	1 0	1 X	1	1
1	0	0 1	X 1	0	1
1	1	X 0	X 0	1	0

QUICK TRICK

Excitation tricks:

D: $D = Q^+$ (just copy what you want).

T: $T = Q \oplus Q^+$ ($T = 1$ only when the state must change).

JK: J decides about going **up** (from 0), K decides about going **down** (from 1). To go $0 \rightarrow 1$ you need $J = 1$ ($K = X$). To go $1 \rightarrow 0$ you need $K = 1$ ($J = X$). To stay at 0 you need $J = 0$ ($K = X$). To stay at 1 you need $K = 0$ ($J = X$).

SR: the same idea, but the other input must be 0 where JK has X for the going-up and going-down cases.

11.4 Race-around condition and master-slave flip-flop

● LEVEL 3 • GATE

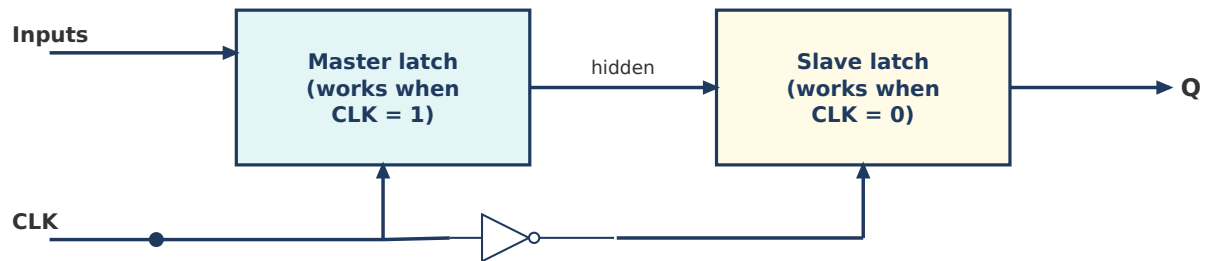
The problem. Take a **level-triggered** JK flip-flop with $J = K = 1$. While the clock stays HIGH, Q toggles. But the new Q is fed back and toggles it **again**, and again, as long as the clock is HIGH (each toggle takes one propagation delay). When the clock falls, the final Q is **unpredictable**. This is called the **race-around condition**.

GATE POINT

Race-around happens **only** when: (1) the flip-flop is **level triggered**, (2) $J = K = 1$ (a T flip-flop with $T = 1$ also), and (3) the clock HIGH time is longer than the propagation delay.

Cures: (1) use **edge triggering**, (2) use a **master-slave** flip-flop, (3) make the clock pulse shorter than the propagation delay (hard to control).

Master-slave idea. Two latches in a row. The **master** works when $CLK = 1$. The **slave** works when $CLK = 0$ (it gets the inverted clock). So the output Q can change only when the clock **falls**, and the output cannot feed back to the master during the same clock level.



QUICK TRICK

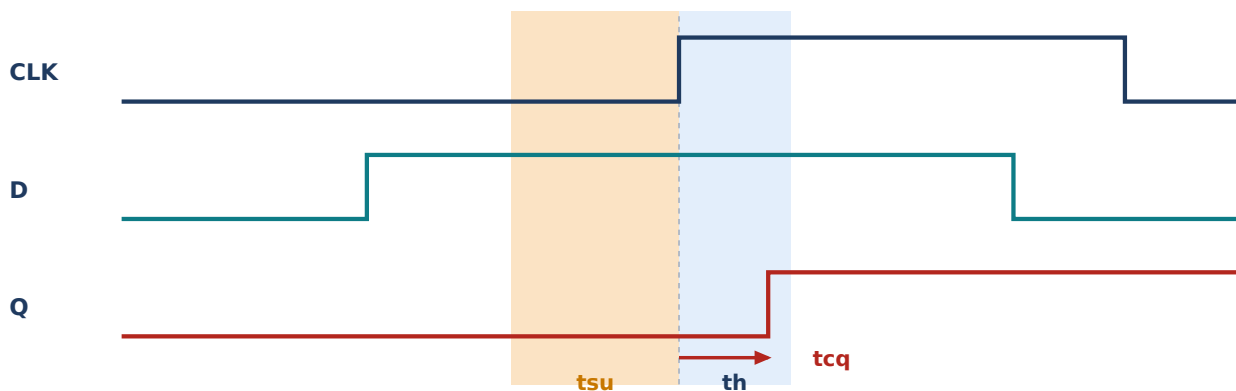
Master-slave = two doors in a corridor. Only one door is open at a time. So the signal can never run through both. That stops the race.

11.5 Setup time, hold time, clock-to-Q delay

● LEVEL 3 · GATE

A flip-flop needs its input to be **steady for a short time around the clock edge**.

Name	Symbol	Meaning
Setup time	t_{su}	The input must be stable BEFORE the active clock edge for at least this long.
Hold time	t_h	The input must stay stable AFTER the clock edge for at least this long.
Clock-to-Q delay	t_{cq}	Time after the clock edge until the output Q actually changes.



QUICK TRICK

Remember: Setup = BEFORE the edge. Hold = AFTER the edge. If D changes inside the shaded window, the flip-flop may store a wrong value (this is called metastability).

Maximum clock frequency. The signal leaves one flip-flop, goes through logic, and must reach the next flip-flop **before** the next edge:

$$\begin{aligned}
 T(\text{clk}) \text{ must be at least: } & t_{cq} + t_{\text{logic}(\text{max})} + t_{su} & f(\text{max}) &= 1 / T(\text{clk}) \\
 \text{Hold check : } & t_{cq} + t_{\text{logic}(\text{min})} \geq t_{\text{hold}}
 \end{aligned}$$

Example (Level 3)

Given: $t_{cq} = 2 \text{ ns}$, logic delay = 6 ns, $t_{su} = 2 \text{ ns}$.

Concept: smallest clock period = $t_{cq} + \text{logic} + t_{su}$.

Calculation: $2 + 6 + 2 = 10 \text{ ns} \rightarrow f_{\max} = 1 / 10 \text{ ns}$.

Answer: 100 MHz.

EXAM TRAP

Use the **maximum** logic delay for the frequency (setup) check. Use the **minimum** logic delay for the hold check. Changing the clock speed can fix a setup problem but **cannot** fix a hold problem.

Quick-Solve Tricks — Summary

Topic	Trick
SR	Set = 1, Reset = 0, 00 hold, 11 illegal. NOR latch invalid at 11, NAND latch invalid at 00.
JK	00 same, 01 $\rightarrow$ 0, 10 $\rightarrow$ 1, 11 $\rightarrow$ opposite (toggle).
D	$Q^+ = D$. Just copy D at the edge.
T	$T = 1$ toggles. Divide by 2.
Excitation	$D = Q^+$. $T = Q \oplus Q^+$. JK: J decides going up, K decides going down.
Timing diagram	Look at the inputs just BEFORE each active edge.
Latch vs FF	Latch = level. Flip-flop = edge.
Race-around	Level-triggered JK with $J = K = 1$ and a long clock. Cure: master-slave or edge-triggering.
Setup / hold	Setup = before the edge. Hold = after the edge.
Max frequency	$T(\text{clk}) \geq t_{cq} + t_{\text{logic}} + t_{su}$

GATE QUICK CHECK — Chapter 11

SR : $Q^+ = S + R'Q$ (SR = 0) JK : $Q^+ = JQ' + K'Q$
D : $Q^+ = D$ T : $Q^+ = T \text{ xor } Q$
Excitation : $D = Q^+$ $T = Q \text{ xor } Q^+$
JK: 0 $\rightarrow$ 1 (1,X) 1 $\rightarrow$ 0 (X,1) 0 $\rightarrow$ 0 (0,X) 1 $\rightarrow$ 1 (X,0)
Max clock frequency = $1 / (t_{cq} + t_{\text{logic}} + t_{su})$

Traps: NOR vs NAND latch invalid state · JK toggle · race-around needs a level-triggered JK with $J = K = 1$ · a T flip-flop with $T = 1$ halves the frequency · hold violation is not cured by a slower clock.

Practice questions

● LEVEL 1 · BEGINNER

Q1. A JK flip-flop has $J = K = 1$ and $Q = 0$. What is Q after the clock edge?

Q2. What happens in an SR flip-flop if $S = R = 1$?

Q3. What is the main difference between a latch and a flip-flop?

● LEVEL 2 · EXAM

Q4. A T flip-flop with $T = 1$ is clocked at 2 MHz. What is the frequency of Q ?

Q5. A JK flip-flop starts with $Q = 0$. (J, K) at four successive clock edges are (1,0), (1,1), (0,1), (1,1). Find Q after each edge.

Q6. A D flip-flop starts with $Q = 0$. $D = 1, 0, 1, 1$ just before four successive active edges. Find Q after each edge.

● **LEVEL 3 • GATE**

Q7. Under which three conditions does race-around occur?

Q8. $t_{cq} = 3$ ns, logic delay = 9 ns, $t_{su} = 3$ ns. Find the maximum clock frequency.

Q9. NAND SR latch: what happens when $S' = 0$ and $R' = 0$?

Q10. Three T flip-flops ($T = 1$) are connected in a chain, each output driving the next clock. The first clock is 16 MHz. Find the frequency at the last output and the number of states.

Q11. Which inputs are needed for a JK flip-flop to change from 1 to 0? From 0 to 1?

Q12. A flip-flop has $t_h = 4$ ns. The previous flip-flop has $t_{cq} = 2$ ns and the logic between them can be as fast as 1 ns. Is there a hold violation?

Answers

Q	Solution
1	$J = K = 1 \rightarrow$ toggle. Q goes from 0 to 1 .
2	Invalid (illegal) input. The outputs are not reliable.
3	A latch is level -triggered. A flip-flop is edge -triggered (it changes only at the clock edge).
4	$T = 1$ toggles at each edge, so the output frequency is half: 1 MHz .
5	(1,0) set $\rightarrow$ 1 . (1,1) toggle $\rightarrow$ 0 . (0,1) reset $\rightarrow$ 0 . (1,1) toggle $\rightarrow$ 1 . Sequence: 1, 0, 0, 1 .
6	Q follows D at each edge: 1, 0, 1, 1 .
7	(1) Level-triggered JK flip-flop, (2) $J = K = 1$, (3) clock HIGH time longer than the propagation delay.
8	Given: 3, 9, 3 ns. Concept: $T = t_{cq} + \text{logic} + t_{su}$. Calculation: $3 + 9 + 3 = 15$ ns $\rightarrow 1/15$ ns. Answer: 66.7 MHz .
9	Invalid condition. Both outputs Q and Q' become 1, and the final state is unpredictable when the inputs go back to 1.
10	Each stage divides by 2: $16 \rightarrow 8 \rightarrow 4 \rightarrow 2$. Output = 2 MHz (divide by $2^3 = 8$). The chain has $2^3 =$ 8 states .
11	$1 \rightarrow 0$: K = 1 (J is X). $0 \rightarrow 1$: J = 1 (K is X).
12	Hold check: $t_{cq} + \text{logic}(\min) = 2 + 1 = 3$ ns, which is less than $t_h = 4$ ns $\rightarrow$ yes, hold violation . (Slowing the clock does not help; the logic delay or clock skew must be changed.)

Next: Chapter 12 — Flip-Flop Conversion (SR $\rightarrow$ JK, JK $\rightarrow$ T, JK $\rightarrow$ D, D $\rightarrow$ T, T $\rightarrow$ D... step by step with the excitation table).